

HDOC 31

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Batch: 16

1. Algorithm

1. Start main program.
2. Read the number of rows and columns from the user.
3. Declare a 2D integer array (matrix) and read its elements using nested loops.
4. Display the operations menu and read the user's choice.
5. Case 1 (Sum of all elements): Loop through all elements, accumulate their sum in a variable, and display the result.
6. Case 2 (Sum of each row): Declare a 1D array of size equal to the number of rows. Loop through each row, calculate the sum of its elements, store it in the 1D array, and print the row sum. Display the contents of the 1D array.
7. Case 3 (Transpose): Use nested loops where the outer loop iterates through columns and the inner loop iterates through rows. Print the matrix elements to display the transpose.
8. Case 4 (Exit): Terminate the loop.
9. Repeat steps 4-8 until the user selects Exit.
10. Stop

2. Test-Case Table

TC ID	Scenario	Inputs	Expected Output
TC-01	Sum of all elements	Rows: 2, Cols: 3, Matrix: [[1, 2, 3], [4, 5, 6]], Choice: 1	Sum of all elements = 21
TC-02	Sum of each row	Choice: 2	Sum of row 1 = 6, Sum of row 2 = 15. Array contains: [6, 15]
TC-03	Transpose of matrix	Choice: 3	Matrix printed as 3 rows and 2 columns.
TC-04	Exit	Choice: 4	Exiting program...

3. C Program Source Code

```
#include <stdio.h>

void sumOfAllElements(int rows, int cols, int matrix[rows][cols]); void sumOfEachRow(int rows,
int cols, int matrix[rows][cols]); void transposeMatrix(int rows, int cols, int matrix[rows][cols]);

int main() {
    int rows, cols, choice;

    printf("Enter the number of rows and columns: "); scanf("%d %d", &rows,
&cols);

    int matrix[rows][cols];

    printf("Enter the matrix elements:\n"); for (int i = 0; i < rows;
i++) {
        for (int j = 0; j < cols; j++) { scanf("%d", &matrix[i][j]);
        }
    }

    do {
        printf("\n--- 2D Array Operations Menu ---\n"); printf("1. Find the sum of all
elements in a matrix\n");
        printf("2. Find the sum of each row of a matrix and store it in an array\n"); printf("3. Find the transpose of a matrix\n");
        printf("4. Exit\n"); printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) { case 1:
            sumOfAllElements(rows, cols, matrix); break;
            case 2:
            sumOfEachRow(rows, cols, matrix); break;
            case 3:
            transposeMatrix(rows, cols, matrix); break;
            case 4:
            printf("Exiting program...\n"); break;
            default:
            printf("Error: Invalid choice.\n");
        }
    } while (choice != 4);

    return 0;
}

void sumOfAllElements(int rows, int cols, int matrix[rows][cols]) { int totalSum = 0;
    for (int i = 0; i < rows; i++) {
```

```

        for (int j = 0; j <
            cols; j++) {
            totalSum +=
                matrix[i][j];
        }
    }
    printf("Sum of all elements = %d\n", totalSum);
}

void sumOfEachRow(int rows, int cols, int
    matrix[rows][cols]) { int rowSumArray[rows];
    printf("Row sums:\n");
    for (int i = 0; i <
        rows; i++) { int sum
            = 0;
            for (int j = 0; j <
                cols; j++) { sum +=
                    matrix[i][j];
            }
            rowSumArray[i] = sum;
            printf("Sum of row %d = %d\n", i + 1, sum);
        }

    printf("The row-sum array
contains: ["); for (int i = 0; i
< rows; i++) {
        printf("%d",
            rowSumArray[i]); if
            (i < rows - 1) {
                printf(", ");
            }
        }
    }
    printf("\n");
}

```

4. Compilation and Execution

```

(kali@kali)-[~/liak]
$ cc hdoc_31.c

(kali@kali)-[~/liak]
$ ./a.out
Enter the number of rows and columns: 2 3
Enter the matrix elements:
1 2 3 4 6 5

— 2D Array Operations Menu —
1. Find the sum of all elements in a matrix
2. Find the sum of each row of a matrix and store it in an array
3. Find the transpose of a matrix
4. Exit
Enter your choice: 1
Sum of all elements = 21

— 2D Array Operations Menu —
1. Find the sum of all elements in a matrix
2. Find the sum of each row of a matrix and store it in an array
3. Find the transpose of a matrix
4. Exit
Enter your choice: 2
Row sums:
Sum of row 1 = 6
Sum of row 2 = 15
The row-sum array contains: [6, 15]

— 2D Array Operations Menu —
1. Find the sum of all elements in a matrix
2. Find the sum of each row of a matrix and store it in an array
3. Find the transpose of a matrix
4. Exit
Enter your choice: 3
Transpose of the matrix:
1      4
2      6
3      5

— 2D Array Operations Menu —
1. Find the sum of all elements in a matrix
2. Find the sum of each row of a matrix and store it in an array
3. Find the transpose of a matrix
4. Exit
Enter your choice: 3
Transpose of the matrix:
1      4
2      6
3      5

— 2D Array Operations Menu —
1. Find the sum of all elements in a matrix
2. Find the sum of each row of a matrix and store it in an array
3. Find the transpose of a matrix
4. Exit

```